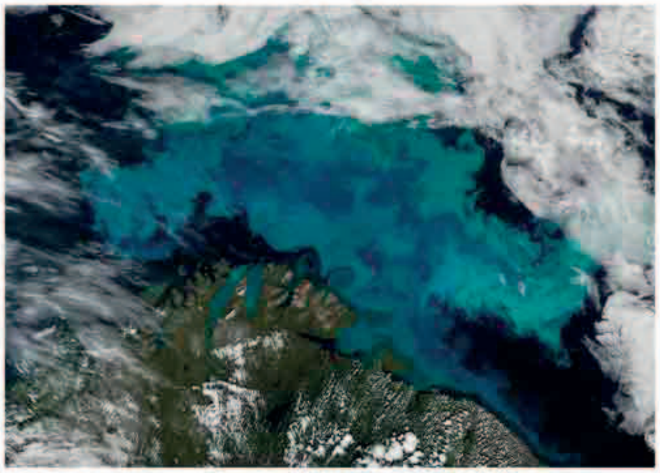
Icy oceans

The open Arctic Ocean and the Southern Ocean around Antarctica freeze over in winter, then melt again in summer. The winter ice can seem almost lifeless but, as it melts, the midnight Sun of the polar summers encourages the growth of drifting, plantlike phytoplankton. These provide food for tiny creatures that multiply fast in the cold, oxygen-rich water, creating a feast for other animals.



LIVING WATERS

In winter, the phytoplankton lie dormant beneath the thick sea ice. As the Sun appears in spring, it thins the ice, so light gets through and the phytoplankton start multiplying. By midsummer, greenish blooms of phytoplankton spread across the oceans. Then, as winter approaches, the Sun sinks from sight and the sea freezes over again.

◂ SATELLITE VIEW
Swirling blooms of summer phytoplankton fill the Barents Sea in the Arctic.

DRIFTING SWARMS

Tiny creatures called zooplankton live under the winter ice alongside the phytoplankton. In spring, the zooplankton start eating the phytoplankton in vast quantities, and are soon breeding. Around Antarctica, small shrimplike krill swarm across huge areas of the Southern Ocean.

▸ TEEMING KRILL
Swarming Antarctic krill swim near the surface, where their massed bodies can make the ocean water appear red.

ANTIFREEZE

Plankton feed fish such as these Arctic cod, and are eaten by all kinds of marine animals, including bottom-living shellfish. Antifreezes in their blood allow these fish and shellfish to survive temperatures that can sink below freezing. Freshwater freezes at 32°F (0°C), but salt makes the freezing point of seawater lower, at about 28°F (-2°C).